Road Damage Detector

Agenda

Manual road inspection is time-consuming, labor-intensive, and often subjective.

Survey teams must physically scan long stretches, leading to slow coverage and frequent oversight.

Resource constraints prevent regular inspections, especially on rural or low-traffic roads

Road damage, if undetected and unrepaired, poses significant risks:

Develop an AI-powered solution for proactive road maintenance:

To design and train machine learning and deep learning models that analyze images or video streams from smartphones, dashcams, or drones to detect and classify road damages in real time.

To provide an automated and scalable monitoring system that covers more kilometers with greater accuracy, enabling early identification of issues.

To seamlessly integrate with authority dashboards, generating actionable insights such as risk scores, severity mapping, and maintenance prioritization.

To ultimately improve road safety, reduce maintenance expenditure, and enhance citizen satisfaction through data-driven, intelligent infrastructure management.

Vision or Mission

To build intelligent, self-updating road networks that ensure safety, efficiency, and sustainability in global transportation systems.
We envision roads that communicate their condition in real time, enabling instant response and predictive maintenance.
By integrating with everyday tools like Google Maps, we aim to empower both governments and travelers with actionable, transparent road health data.

Our mission is to design and deploy scalable, intelligent systems that:

1. Integrate with Google Maps to help users:

- Identify safest travel routes based on road condition.
- Avoid damaged zones in real time.
- Report new damage via a “Road Condition” layer.
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1. Reduce accidents, lower repair costs, and create a more efficient and safer transport ecosystem.

1. Deliver automated, geo-tagged damage reports to municipal dashboards.

Our Roadmap

1. Data collection and annotation
2. AI model development and testing
3. Pilot deployment with city partners
4. Scale-up and integration with public dashboards

Who Faces the Problem and Why does It Matter

Timely detection of road damage prevents minor issues from escalating into major hazards, protecting both lives and infrastructure. By enabling proactive repairs, it ensures smoother traffic flow and boosts public confidence in road safety.

🚗 Vehicle Owners & Commuters

1.) Suffer vehicle damage, delays, and fuel waste
2.) ₹5,000–₹30,000 per repair
3.) 72% encounter faults weekly

🏍️ Vulnerable Road Users

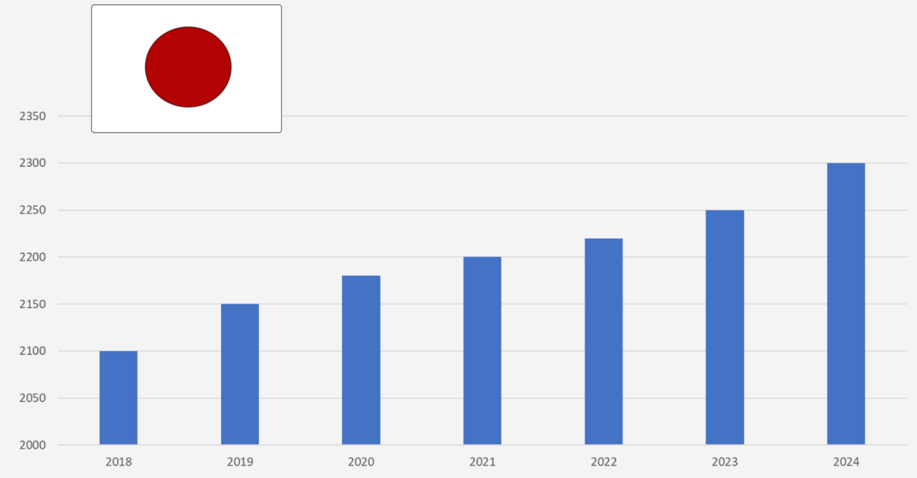
1) Two-wheelers, cyclists, pedestrians
2) Greater risk for serious injury
3) Particularly dangerous in low-visibility areas

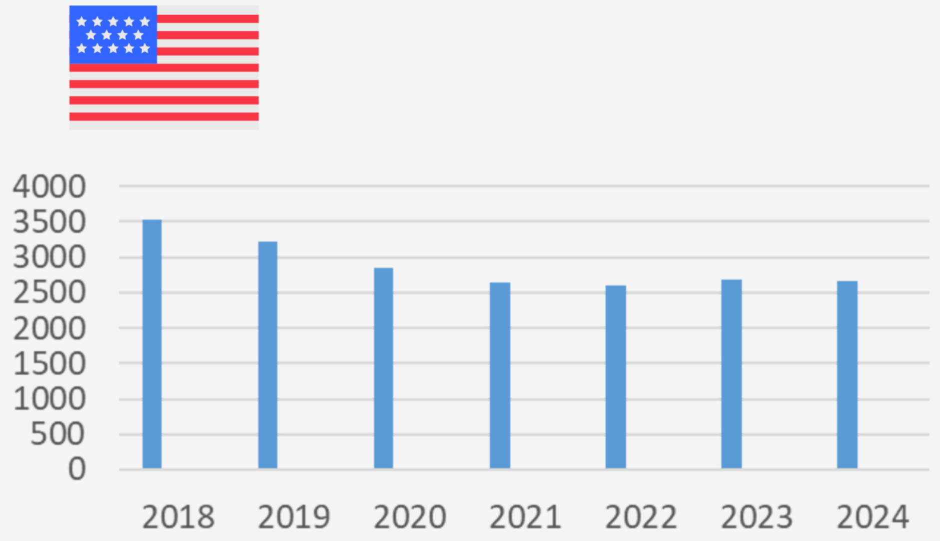
🏛️ Authorities & Planners

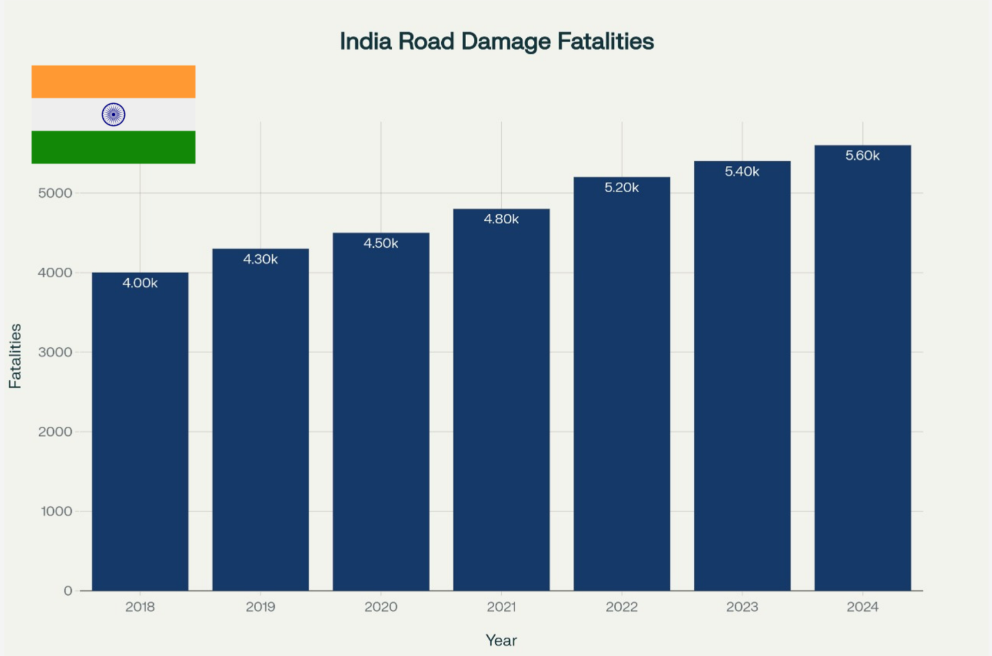
1) Rising maintenance and legal costs
2) Emergency repairs vs. proactive fixes
3) Increased complaint volume

💰 Economic & Societal Impact

1).Supply chain & business disruptions
2) Increased logistics costs
3) Lower quality of life, more emissions







Challenges

Building a robust dataset for road damage detection is challenging due to the need to capture many damage types, road conditions, and environments.

High-quality images are essential, as issues like blur, shadows, or poor lighting reduce AI accuracy.

Limited or poorly labeled data further lowers detection performance, especially in difficult conditions.

Integrating road damage detection with maps faces key challenges, including inaccurate GPS alignment (causing errors in pinpointing damage), difficulties in efficiently visualizing many detections in real time, and dealing with API rate limits and custom overlay restrictions of mapping services.

Live integration can suffer from synchronization delays, leading to outdated map data, and handling sensitive location-linked data requires strict privacy controls and regulatory compliance.

To build a robust dataset for road damage detection, collect diverse images from smartphones, dashcams and drones across various conditions.

Filter and enhance image quality to ensure clarity, and apply accurate labeling with data augmentation to improve model reliability in challenging environments.

To improve map integration, enhance GPS accuracy using sensor fusion and map-matching algorithms. Use clustering and heatmaps for clear real-time visualization while optimizing API calls via batching and caching. Employ real-time data push technologies to minimize delays, and ensure privacy by anonymizing location data and complying with regulations.

Marking Damage on Map

- Each identified damage is visualized as a red marker on the map interface.
- Markers help in quick visual identification and tracking of damage points.

🕹️Interactive Features:

- Facilitates quick inspection, prioritization, and planning of repair actions.
- Clicking a marker opens a popup window with complete details.

🧠 Details Captured per Marker:

Type of Damage

- Specifies the nature of damage (e.g., surface crack, deep pothole, erosion).

Confidence Level

- A percentage score reflecting how confident the model is about the classification.

Timestamp

- Records when the damage was first detected—helps track changes over time.

Pothole Detection & Reporting

- **Multi-Method Reporting:** Report potholes via Image Upload or Live Webcam.
- **AI-Powered Detection:** Uses a YOLOv8 model to accurately identify potholes.
- **Smart Data Capture:** Captures Geolocation (Lat/Lon), user feedback, and ratings.
- **Intelligent Analysis:**
- **Calculates Danger Percentage based on pothole size.**
- **Checks for Duplicate Potholes within a 15m radius to avoid redundant entries.**

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Pothole Lifecycle Management

- Role-Based Access: Secure login for Admin, Maintenance, & Public users.
- Live Status Tracking: Monitor the entire workflow from 'Reported' -> 'Under Construction' -> 'Repaired'.
- Task Assignment: Admins can view all reports and assign tasks to maintenance teams.

Key Features Overview

🔐 User Authentication & Authorization

- Secure registration with bcrypt password hashing
- Role-based access (e.g., admin, maintenance, user)
- Session management using Flask-Login
- Login required for protected routes

🕳 Pothole Detection & Reporting

- Upload image with geolocation (lat/lon)
- YOLOv8 model detects potholes, returns annotated image
- Checks for duplicate potholes within 15m radius
- Severity classification (Low, Medium, High)
- Road Quality Score (RQS) based on pothole area

🔄 Pothole Lifecycle Management

- View all reported potholes with status and location
- Admins/maintenance can update pothole status
- Users can give feedback via comments and ratings

📡 Real-Time Detection Features

- live_detection page planned for webcam-based detection
- Handles webcam access errors gracefully

⚙️ Tech Stack & Infrastructure

- Built with Flask, SQLAlchemy, Bcrypt, Flask-Login
- PostgreSQL database with normalized tables
- Uses .env for secure configs (SECRET_KEY, DATABASE_URL)

Meet The Team

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